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A Life Cycle Approach to Subsurface Water Disposal Design for Unconventional Gas Development

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Abstract

Effective management of subsurface water disposal is critical for unconventional gas development due to the large volumes of water flowed-back and produced. This study introduces a comprehensive, life cycle-based framework for subsurface water disposal focusing on unconventional gas development.

The framework combines subsurface characterization, injectivity testing, and environmental impact assessment. Its approach also integrates field data, reservoir simulation, and geochemical and economic modeling with long-term risk mitigation and monitoring.

This study demonstrates that integrating a life cycle perspective into water disposal design enhances decision-making by balancing technical feasibility, economic viability, and reduces the environmental impacts associated. Key findings indicate that identification of optimal injection method and proper reservoir characterization reduces risks of formation overpressure and groundwater contamination. Additionally, long-term monitoring of disposal wells ensures operational safety and regulatory compliance. The findings also highlight the necessity of proactive risk management which can enable us to develop a customized disposal strategy that balances operational demands with environmental and regulatory considerations.

Introduction

Unconventional gas development—including tight gas, and shale gas—produces significant volumes of water as a byproduct of hydraulic fracturing (Figure 1). Managing this produced and flowed-back water in a safe, cost-effective, and environmentally sustainable manner is a critical challenge for operators. The disposal of such water typically involves subsurface injections, which present multiple risks if not carefully managed, including over pressuring, induced seismicity, and contamination of clean aquifers (Veil, 2009; Nicot et al., 2014).

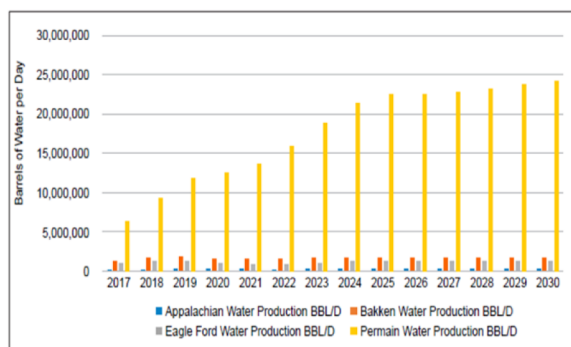


Figure 1—Produced Water Volumes in Oil and Gas Development Regions. Source: U.S. Department of Energy. (2024, June 18).

The scale of water production in modern shale plays has grown considerably, driven by extended lateral lengths, increased proppant loading, and high-volume stimulation designs. As such, water management is now a defining factor in the economics and environmental footprint of unconventional operations. Traditional disposal practices often focus on short-term logistical and financial considerations, prioritizing immediate disposal capacity over long-term geological sustainability. For example, increased seismicity has been linked to saltwater disposal in Oklahoma (Figure 2) (Walsh and Zoback, 2015).

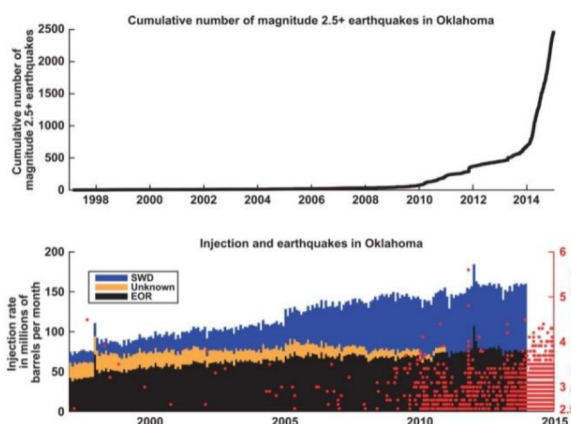


Figure 2—(Top) Cumulative number of M2.5 or greater earthquakes in Oklahoma since 1997. (Bottom) The left axis shows the total combined injection rate of all Underground Injection Control (UIC) wells in Oklahoma by type (see the text). The right axis shows all earthquakes in the state by magnitude through time in the state. Earthquake data are completed through 2014. The injection data are only available throughout 2013. Source: Walsh, F. R., & Zoback, M. D. (2015).

Moreover, regulatory agencies and stakeholders are increasingly emphasizing the need for transparent reporting, improved risk management, and alignment with Environmental, Social, and Governance (ESG) criteria. In this context, decision-making frameworks must evolve beyond compliance checklists toward integrated models that account for the full life cycle of disposal assets. Digital tools like reservoir simulation and machine learning (ML) are now being used to improve how disposal wells are located, sized, and managed.

This paper presents a life cycle-based strategy that incorporates reservoir simulation, environmental assessment, economic evaluation, and injectivity testing. It proposes a sustainable design approach rooted in Life Cycle Assessment (LCA) and supports adaptive decision-making throughout the life of a disposal well. By integrating technical, environmental, and economic disciplines into a single framework, the proposed methodology enables operators to proactively mitigate risk, optimize performance, and comply with evolving regulatory standards. The approach also facilitates the identification of subsurface constraints and the evaluation of long-term injectivity trends, which are critical for ensuring operational continuity and environmental protection in unconventional gas developments.

Motivation and Challenges

Cumulative disposal water burden stresses local aquifers, surface treatment infrastructure, and most critically, the subsurface disposal formations.

Key challenges include:

- **Geological Uncertainty:** Many unconventional formations exhibit significant lateral heterogeneity, complicating the selection of disposal zones.
- **Regulatory Pressure:** New environmental legislation and public examination require detailed documentation of disposal risks and mitigation strategies.
- **Environmental Risks:** Induced seismicity, brine migration, and aquifer contamination are increasingly associated with high-volume disposal wells, particularly in regions such as Oklahoma and West Texas ([Walsh and Zoback, 2015](#)).
- **Economic Constraints:** High operational and capital expenditures (OPEX/CAPEX) necessitate optimized well and facility designs to maintain profitability.

Conventional vs. LCA Approach

Conventional Design Approaches

Traditional subsurface water disposal systems are often driven by immediate operational convenience and cost constraints. Normally, the conventional design approach is to drill disposal wells close to production pads to reduce costs. Injection operations are conducted at moderate rates, assuming the formation's capacity will accommodate ongoing water volumes without significant performance degradation.

However, such simplified approaches often lack robust geo-mechanical and geochemical evaluation. They typically assume linear injectivity profiles, overlook spatial heterogeneity in reservoir permeability and pressure, and fail to anticipate long-term consequences like formation damage or induced seismicity. Over time, this can result in rapid injectivity decline, increased surface pressure requirements, wellbore integrity issues, or fracture growth beyond the intended disposal zone—potentially compromising rock integrity and regulatory compliance.

Moreover, disposal wells that are poorly located can disrupt nearby production or storage operations, allow pressure to transfer through fault zones, or increase the risk of contaminating protected aquifers if the sealing rock layers are not properly assessed. These issues highlight the importance of adopting more integrated and forward-looking disposal planning.

Life Cycle-Based Design Framework

A life cycle-based disposal strategy marks a major shift from short-term, reactive decisions to long-term, proactive planning. Instead of only focusing on immediate needs, this approach looks at the entire lifespan of a disposal system—from design and operation to closure and monitoring. It brings together different types of data early in the planning stage, such as rock properties, fluid behavior, chemical composition, and cost factors, to help predict how the system will perform over time.

Life Cycle Workflow for subsurface Water Disposal Design

Designing safe and efficient water disposal systems for unconventional oil and gas operations requires a step-by-step approach that looks at the entire life cycle of the project. This helps ensure that each phase—from early planning to long-term monitoring—is done correctly, cost-effectively, and with care for the environment. The process includes several elements as listed below:

- Subsurface Characterization and Reservoir Modeling.

- Geochemical Considerations in Disposal Design.
- Injectivity Testing.
- Effects of Design on an Environmental and Economical Perspective.
- Operational Monitoring and Long-Term Planning.

By using Life Cycle Assessment (LCA), operators can also compare the environmental impact of different disposal options. For example, they can evaluate whether centralized disposal hubs, smaller local wells, or recycling systems make the most sense based on energy use, emissions, and water reuse potential. This is especially valuable in unconventional oil and gas fields, where water management is complex and costly.

Overall, this framework encourages flexible and ongoing decision-making, where field data, models, and monitoring results are continuously used to adjust and improve operations.

Subsurface Characterization and Reservoir Modeling

Detailed subsurface characterization is essential to ensure suitable formation selection and to prevent unintended fracturing or fluid migration.

Data sources include:

- **Geophysical Logs:** Provide porosity, permeability, and lithological data.
- **Core Samples and Cuttings:** Identify mineral reactivity and potential for scaling or precipitation.
- **Formation Pressure Data:** Used to calculate the safe operating envelope below the fracture gradient.

Reservoir simulation using various tools supports sensitivity analysis for injection scenarios. For example, modeling can predict how cumulative injection volumes influence formation pressure and identify when risk thresholds—such as formation parting pressure (FPP)—are approached.

By coupling simulation models with historical injection data and falloff test results, reservoir behavior can be predicted over 10 to 30-year timescales, providing a more robust basis for disposal infrastructure planning.

Geochemical Considerations in Disposal Design

The chemical compatibility between the water being injected and the underground rock formation is very important for keeping disposal wells working properly and avoiding damage.

Some common issues include:

- **Scale Buildup:** When injection water mixes with formation water, minerals like barium, strontium, or calcium sulfates can form solid deposits that clog the rock.
- **Clay Swelling:** Certain clays in shale or silty rocks can expand when exposed to injection water, which can block fluid flow.
- **Corrosion and Sour Gas:** Chemical reactions between the injected water and minerals—or underground microbes—can lead to pipe corrosion or the production of harmful gases like hydrogen sulfide.

Life-Cycle water disposal considerations include data-driven forward modeling, to evaluate the likelihood and intensity of issues related to compatibility. This will help guide strategies such as treating the water before injection or preparing the formation with a special flush.

Geochemical modeling also helps choose the right chemical additives (like scale inhibitors), plan when to use treatments, and select formations that are less likely to react with the injected fluids.

Injectivity Testing

Injectivity testing plays an important role in developing and managing water disposal wells, as it helps in accessing a well's ability to safely and effectively take in and inject fluids into a reservoir. This section describes the methods utilized in injectivity testing, the relevant equations, and how testing influences well and field design.

Injectivity Testing Methodologies:

Injectivity testing includes various methods to examine the performance and effectiveness of water disposal wells. Step-rate and falloff tests are primarily the most utilized among them. These assessments provide vital information regarding the injectivity index of the well, as well as the transmissibility of the reservoir, which are crucial for enhancing well design and guaranteeing efficient and safe injection operations.

Step-Rate Test. The step-rate test involves increasing the injection rate in incremental steps and equal durations while observing the pressure response of the well. (Figure 3) (RRC, 2024). It is typically used to estimate (i) Formation Parting Pressure and (ii) rate-dependent skin.

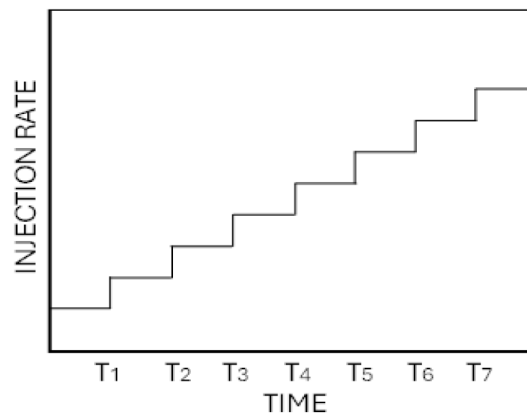


Figure 3—Example of a step-rate test Source: Salazar, A., & Kumar, A. (1992).

The process generally includes these stages:

- **Initial Injection Rate:** Starting with a low injection rate and maintaining it for a specified duration to stabilize the system.
- **Pressure Monitoring:** Continuously monitoring and recording the wellhead pressure during each injection rate.
- **Stepwise Increase:** Gradually increasing the injection rate in predefined increments (e.g., 2000 bbl/day, 3000 bbl/day) while maintaining each rate long enough to achieve a stable pressure response.
- **Data Collection:** Record the injection rate and corresponding pressure for each step.

The injectivity index (II) is calculated using the following equation:

$$II = \frac{q}{\Delta p} = \frac{q_w}{P_{wfi} - P_e}$$

Refer to the Nomenclature section for more details.

Formation Parting Pressure (FPP). Formation Parting Pressure (FPP) refers to the level of pressure at which a reservoir formation begins to fracture. Economides, M. J., & Nolte, K. G. (2000) It is essential for maintaining safe and effective water disposal well injection processes. Surpassing this pressure can result

in unwanted fractures, jeopardizing the well's structural integrity and possibly leading to environmental concerns.

Salazar et al. (1992) provide an in-depth analysis of FPP, offering valuable information about how it is measured and its impact on the design and management of injection wells. Their work highlights the importance of accurately determining FPP to prevent formation damage and maintain well's integrity. By employing step-rate and falloff tests, operators can identify the FPP and adjust their injection strategies, accordingly, ensuring safe and efficient operations.

Understanding FPP allows operators to design wells and injection strategies that avoid exceeding this pressure.

Falloff Test. The falloff test includes injecting fluid at a steady flow rate, the well is then shut in, and the pressure drop is observed. Gu, W., Wu, J., & Sun, Z. (2024).

The procedure typically involves the following steps:

- **Constant Injection Rate:** Injecting fluid at a constant rate for a specified duration to stabilize the system.
- **Shut-In Period:** Closing the well and monitoring the pressure decline over time.
- **Pressure Monitoring:** Continuously recording the pressure during the shut-in period, preferably using high-frequency gauges.
- **Data Analysis:** Using pressure decline data and pressure transient analysis to estimate reservoir's transmissibility and connectivity.

The transmissibility (T) of the reservoir is estimated using the following equation:

$$T = \frac{kh}{\mu}$$

Refer to the Nomenclature section for more details.

Effects of Design on an Environmental and Economical Perspective

Economic Impacts of Injectivity on Well and Field Design. Injectivity testing plays a critical role in the economic optimization of water disposal systems in unconventional resource plays. The injectivity index, determined through step-rate and falloff tests, provides insights into a reservoir's ability to accept fluids efficiently and safely. High injectivity values can justify larger wellbore diameters or centralized disposal sites capable of handling higher flow rates, thereby reducing the need for multiple lower-capacity wells.

Fewer high-performance wells lead to significant capital (CAPEX) and operational (OPEX) savings by:

- Reducing the total number of wells needed.
- Minimizing surface infrastructure such as well pads, pipelines, and access roads.
- Enhancing reliability by concentrating on operations at fewer, more manageable sites.

Conversely, low injectivity results may indicate the need for stimulation or design modifications to avoid excessive reservoir pressure buildup and ensure regulatory compliance.

Optimizing well count and spacing based on injectivity also has a direct impact on long-term monitoring and maintenance costs. With fewer active sites, companies can:

- Simplify operational logistics.
- Reduce environmental monitoring frequency and extent.
- Improve scalability and flexibility for future development phases.

In summary, injectivity testing is not only essential for technical design and regulatory safety but also a powerful driver of economic efficiency. By integrating injectivity data into early field development planning, operators can improve system performance, reduce costs, and limit environmental disturbance.

Surface Footprint and Environmental Risk. Designing disposal systems using data on how easily fluids can be injected into underground formations (injectivity) helps reduce the total number of wells needed. Clark, C. & Veil, J. (2009).

Fewer wells mean:

- Less land disturbance and less damage to wildlife habitats.
- Fewer roads, well pads, and pipelines on the surface.
- A lighter regulatory and permitting load for operators.

By focusing on areas with higher injectivity, companies can reduce their environmental footprint while still meeting disposal needs.

Regulatory and Safety Buffering. Designing disposal systems with an understanding of how much pressure the formation can safely handle helps avoid overloading the rock. By staying 15–20% below the maximum safe pressure, companies can protect the integrity of the formation and stay in line with regulatory requirements EPA (2023).

Operational Monitoring and Long-Term Planning

The current disposal system should be designed in a way that can be adjusted to changing conditions in real time. New technologies are helping operators monitor underground activity more closely and respond quickly when needed.

For example:

- **Pressure testing** between wells shows how connected different parts of the formation are.
- **Fiber Optics, Distributed Acoustic Sensing (DAS)** can detect small earthquakes and possible leaks using fiber optic cables. (Figure 4)
- **Chemical tracers** help track how injected water moves underground and spot unwanted flow between layers.

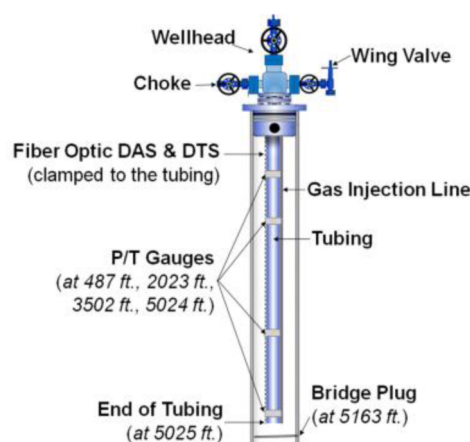


Figure 4—Simplified test-well schematic highlighting the location of the downhole pressure and temperature (P/T) gauges Source: Sharma, J., Cuny, T., Ogunsanwo, T., & Santos, O. (2020).

In addition, machine learning (ML) tools can analyze data like pressure, injection rates, and seismic activity to spot problems early or predict pressure build-up.

This makes it possible to:

- Make real-time changes to how and where water is injected.
- Take quick action, such as reducing flow rates or shutting off certain zones.
- Protect the system and respond faster to regulatory concerns.

Conclusion and Recommendations

Taking a full life cycle approach to underground water disposal helps companies manage technical, environmental, and financial risks more effectively.

To build smarter and more responsible disposal systems, operators should consider the following:

- **Test how easily water can be injected** into the formation to design fewer, safer wells and avoid putting too much pressure underground.
- **Use life cycle assessments (LCA)** early on to understand the pros and cons of different design choices—not just in cost, but also in environmental impact.
- **Adopt real-time monitoring and smart data tools** (like machine learning - ML) to adjust injection operations as conditions change.
- **Support environmental, social, and governance (ESG) goals** by limiting land use, ensuring long-term well safety, and staying compliant with environmental rules.

Nomenclature

II	Injectivity Index (bbl/day/psi)
q	Injection Rate (bbl/day)
Δp	Pressure Differential (psi)
q_w	Water Injection Rate (STB/D)
P_{wfi}	Bottomhole Pressure (psi)
P_e	Reservoir Pressure (psi)
T	Transmissibility (D.ft or mD.ft)
k	Permeability (Darcy D or millidarcy mD)
h	Thickness (ft)
μ	Viscosity (Centipoise cp) bbl/day Barrels per Day psi Pounds per Square Inch
ML	Machine Learning
FPP	Formation Parting Pressure (psi)
P/T	Pressure and Temperature
EPA	Environmental Protection Agency
UIC	Underground Injection Control
LCA	Life Cycle Assessment
CAPEX	Capital Expenditures
OPEX	Operational Expenditure
ESG	Environmental, Social, and Governance
DAS	Distributed Acoustic Sensing

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